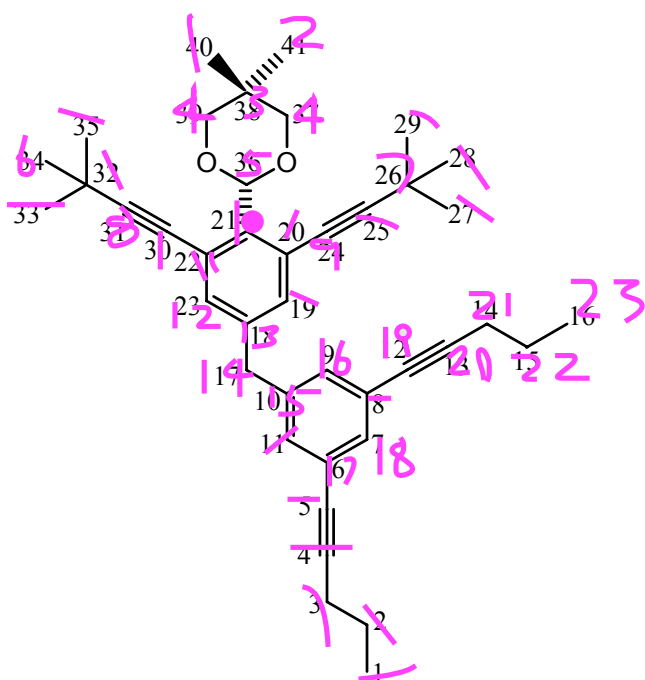
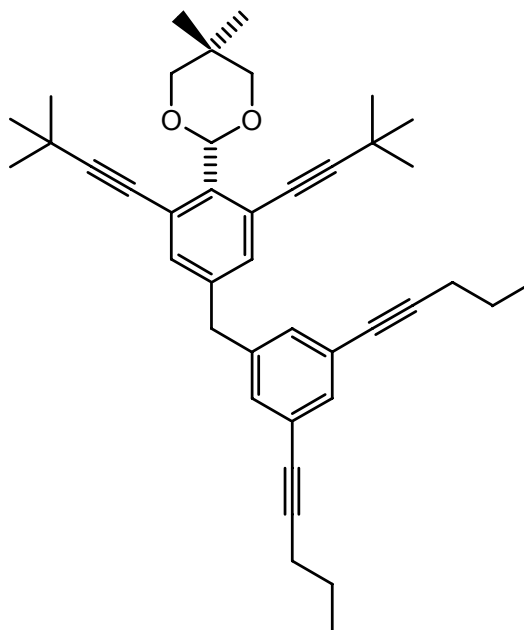


4. This question is about the NMR spectra of NanoPutians

In June 2003, a research paper was published announcing the synthesis of the smallest representations of the human form: 2 nm tall anthropomorphic molecules, nicknamed 'NanoPutians' by their creators.

The molecules synthesised included 'NanoKid', 'NanoBaker' and 'NanoAthlete'. The compound shown to the right was called 'NanoBalletDancer' and has the formula $C_{41}H_{50}O_2$.



When assigning an NMR spectrum, the first step is to identify how many atoms there are in unique environments.

Both carbon atoms (^{13}C) and hydrogen atoms (1H) give NMR signals. Each atom *in a different environment* will give rise to one signal.

For example, in the structure of NanoBalletDancer, carbon atoms 37 and 39 are equivalent; we may write ($37 \equiv 39$). Hence, although there are two carbon atoms (37 and 39) which have one oxygen atom attached, only *one* signal would be observed in a ^{13}C NMR spectrum due to these carbon atoms since they are equivalent.

- (a) Which carbon atoms making up the benzene rings are equivalent? Write down $w \equiv x$, $y \equiv z$ etc for any equivalent atoms. How many signals *in total* would be observed due to benzene-ring carbons in a carbon NMR spectrum of NanoBalletDancer?

22=20, 23=19, 11=9, 6=8
in total, 8

- (b) List the groups of equivalent triple bond carbons in NanoBalletDancer. How many signals would be seen *in total* in the ^{13}C spectrum due to triple bond carbon atoms?

30#31, 24#25, 12#13, 5#4, 4=13, 5=12, 25=31, 24=30, 4 signals total

- (c) How many different methyl groups ($-CH_3$ groups) are there in NanoBalletDancer? Again, list them in groups.

33, 34, 35, 27, 28, 29 is one group. 16, 1 is another group, 40 and 41 are unique

- (d) How many different carbon environments are there in NanoBalletDancer – i.e. how many signals would be seen in total in the ^{13}C NMR spectrum? **23**

Similarly, in ^1H NMR, the total number of signals depends on the number of different environments of hydrogen atoms in a structure. There are 13 different environments of hydrogens in NanoBalletDancer; their signals are labelled **A–M** in the spectrum below. The numbers of hydrogen atoms in each unique environment is given under the label. Hydrogen atoms in similar environments all have similar chemical shifts. For example, all the hydrogens on the benzene rings occur in the same region of the spectrum, i.e. they have a similar *chemical shift*.

However, ^1H NMR is complicated by *coupling*. If a hydrogen is within three bonds of another hydrogen *which is in a different environment*, instead of appearing as a single peak, its signal is split into a number of peaks. In general, if the hydrogen under consideration is within three bonds of n hydrogens in a different environment from the one under consideration, it will be split into $(n + 1)$ peaks. The ratio of the area under the peaks is given by Pascal's triangle as outlined below.

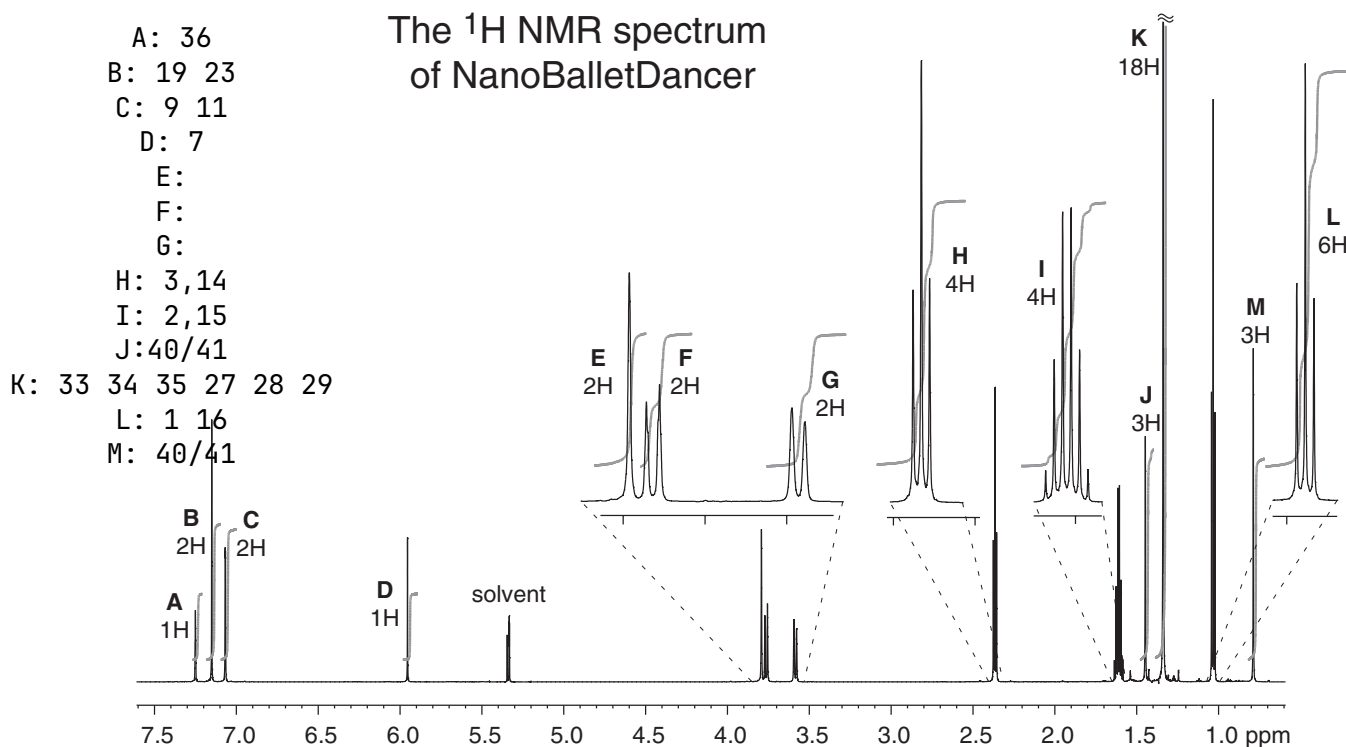
observed ratio	for a hydrogen coupling to
1	← 0 hydrogens
1 : 1	← 1 hydrogen
1 : 2 : 1	← 2 hydrogens
1 : 3 : 3 : 1	← 3 hydrogens
1 : 4 : 6 : 4 : 1	← 4 hydrogens

The signal for a given hydrogen is not split by any hydrogens *which are in the same environment* as it is in.

- (e) Into how many peaks will the signal from a hydrogen that couples with 5 other hydrogens be split? What will the ratio of the peaks be? **6 peaks, 1:5:10:10:5:1**

It is possible to assign the ^1H NMR spectrum of NanoBalletDancer by considering the numbers of hydrogens in different environments, their chemical shifts, and their coupling patterns. For example, the signal at 7.15 ppm (**B**) is due to the hydrogen atoms on carbons 19 and 23.

- (f) On the table in your answer sheet, assign (as far as possible) which signals are due to which hydrogen atoms. The assignment for signal **B** has already been filled in on the answer sheet. (For some signals, it might not be possible to decide between two alternative assignments – in which case just write '... or ...' on the answer sheet.)



(e) Draw the structure of compounds **C** and **D**.

Compound **E** can be hydrolysed to levulinic acid and by-product compound **F**.

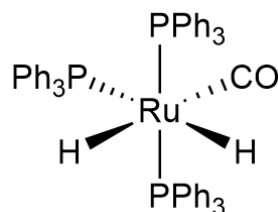
(f) Draw the structure of compound **F**.

The hydrogenation of levulinic acid gives a green and sustainable route to several different chemicals used in perfumes and flavouring. One catalyst for this hydrogenation is the octahedral complex of formula $[\text{RuH}_2(\text{CO})(\text{PPh}_3)_3]$. This complex is named dihydridocarbonyltris(triphenylphosphine)ruthenium(**Z**), where **Z** is the oxidation state of ruthenium.

Text

(g) State the value of **Z**.

The stereochemistry of the catalyst $[\text{RuH}_2(\text{CO})(\text{PPh}_3)_3]$ is shown.

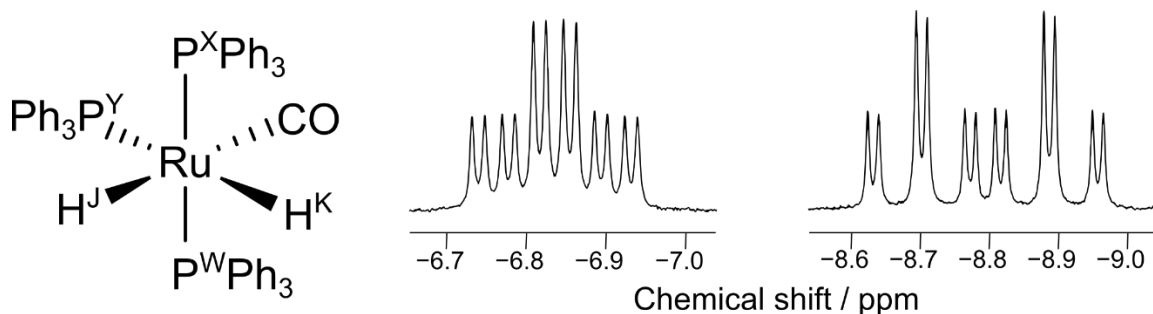


(h) Draw the other stereoisomers of this complex. Tick to indicate whether the stereoisomer shown and each stereoisomer you have drawn has an enantiomer. For a pair of enantiomers, only draw one of the pair.

This catalyst can be analysed by ^{31}P NMR. ^{31}P is 100% abundant and is a spin $\frac{1}{2}$ nucleus like ^1H . The ^{31}P NMR is run in a way that means no coupling between ^{31}P and ^1H is observed. The only coupling which is observed is between ^{31}P nuclei in different environments.

(i) What is observed in the ^{31}P NMR spectrum of this catalyst? Tick the correct answer.

In the ^1H NMR spectrum, the two hydrides appear at negative chemical shifts and have complex coupling patterns. The couplings observed for these signals arise from coupling through two bonds to either ^1H or ^{31}P , (i.e. either $^1\text{H}-\text{Ru}-^1\text{H}$, or $^1\text{H}-\text{Ru}-^{31}\text{P}$) and are measured in Hz. The coupling constant between two nuclei that are *trans* is usually larger than between two nuclei that are *cis*.



The signal at -6.83 ppm is a 'triplet of doublets of doublets'. The signal is split into a triplet with the largest coupling constant of 31 Hz, further split into a doublet with coupling constant of 15 Hz, further split into a doublet with coupling constant of 6 Hz.

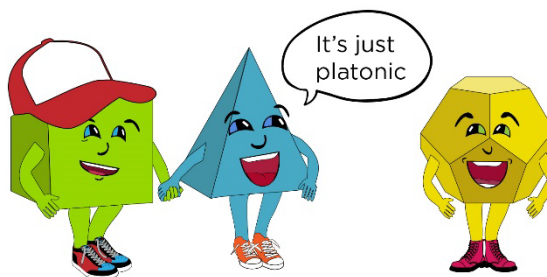
The signal at -8.80 ppm is a 'doublet of triplets of doublets'. The signal is split into a doublet with the largest coupling constant of 74 Hz, further split into a triplet with coupling constant of 28 Hz, further split into a doublet with coupling constant of 6 Hz.

(j) Complete the table to assign the values of the coupling constants in Hz to the following pairs of nuclei ($\text{H}^{\text{J}}-\text{H}^{\text{K}}$, $\text{H}^{\text{J}}-\text{P}^{\text{W}}$, $\text{H}^{\text{J}}-\text{P}^{\text{X}}$, $\text{H}^{\text{J}}-\text{P}^{\text{Y}}$, $\text{H}^{\text{K}}-\text{P}^{\text{W}}$, $\text{H}^{\text{K}}-\text{P}^{\text{X}}$, $\text{H}^{\text{K}}-\text{P}^{\text{Y}}$).

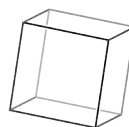
3. This question is about cubane

Platonic solids are three-dimensional shapes where all faces, edges, and angles are congruent (identical). There are only five such solids. Plato hypothesised that the classical elements — air, earth, fire, and water — each corresponded to one of the shapes.

With the advances in chemistry in the last 2360 years, we no longer think that there are four elements or that they are made of Platonic solids. However, we can make a Platonic solid out of elements.



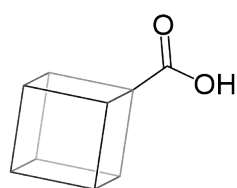
One such molecule, cubane, is of interest and potential use because of its unusual structure.



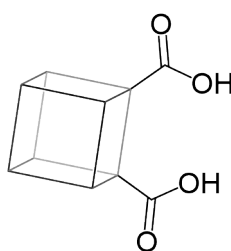
Skeletal formula of cubane

(a) State the molecular formula of cubane.

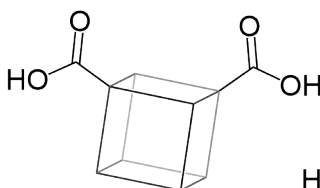
One synthesis of cubane proceeds via cubane-1,4-dicarboxylic acid. The structures of cubane-carboxylic acid and the three isomers of cubane-dicarboxylic acid are shown below.



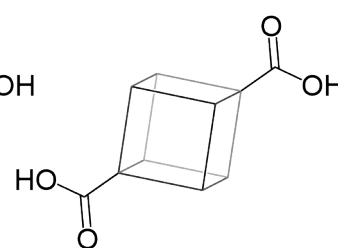
Cubane-carboxylic acid



Cubane-1,2-dicarboxylic acid



Cubane-1,3-dicarboxylic acid



Cubane-1,4-dicarboxylic acid

(b) Complete the table in the answer booklet with the number of peaks in the ^{13}C NMR spectrum of:

- (i) Cubane
- (ii) Cubane-carboxylic acid
- (iii) Cubane-1,2-dicarboxylic acid
- (iv) Cubane-1,3-dicarboxylic acid
- (v) Cubane-1,4-dicarboxylic acid.